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The Editors
Physical Chemistry Chemical Physics

Dear Editors,

On behalf of my co-author, I am pleased to submit our manuscript, **“Ultrafast recombination and exchange coupling preclude weak-field magnetosensitivity of neurotransmitter-metabolising flavoenzymes,”** for consideration in *Physical Chemistry Chemical Physics*.

The manuscript sits squarely in the spin-chemistry scope of PCCP. Using an exact Liouville-space radical-pair master equation that includes spin-selective recombination, exchange and dipolar coupling, hyperfine interactions, and electronic decoherence, we establish a quantitative bound on whether a weak, static magnetic field can bias the catalytic flux of the neurotransmitter-metabolising flavoenzymes (monoamine oxidase A and B, D-amino-acid oxidase) through the radical-pair mechanism. Two independent active-site properties—a gigahertz exchange coupling and a picosecond recombination lifetime—each suppress the geomagnetic-field effect below the numerical resolution of the solver, and we show the null is robust to the form of the electronic decoherence (including the singlet–triplet dephasing channel appropriate to a tightly bound pair) and to the quantum Zeno mechanism. Cryptochrome, treated identically, reproduces the expected percent-level sensitivity as a positive control. The result quantifies a methodological boundary of broad relevance to the community: an RPM estimate for an active-site cofactor radical must propagate the physical lifetime and electronic decoherence, or it overestimates the weak-field effect by orders of magnitude. The prediction is consistent with the one experiment to scan static fields on brain monoamine oxidase, which found no change from 0.1 to 340 mT.

This is a complete study, developed independently and written from scratch around the master-equation formalism; it is not under consideration elsewhere and has no prior publication. All code is openly released and the central claims are covered by an automated test suite. The authors declare no competing interests and the study received no specific funding. We suggest the manuscript is well matched to referees working in radical-pair spin dynamics and quantum biology.

Thank you for considering our submission.

Yours sincerely,

Hikaru Wakaura
on behalf of both authors